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LAB 6: SWITCH COMMUNICATION BETWEEN TWO LAN

1. OBJECTIVE

To interconnect two Layer-2 switches in a LAN, connect six PCs across them, and verify end-to-end communication using ICMP in Cisco Packet Tracer Simulation Mode.

2. APPARATUS

Item	Specification	Qty
Switch	Cisco 2960	2
PC	Generic (Packet Tracer)	6
Cable	Copper Straight-Through	7
Software	Cisco Packet Tracer	1

3. THEORY

A switch forwards frames within a LAN using MAC address tables. Two interconnected switches form a single broadcast domain — PCs on different switches communicate freely within the same IP subnet. ICMP ping confirms reachability. STP (802.1D) runs automatically to prevent loops; STP events in the simulation are normal behavior.

4. NETWORK TOPOLOGY

Connection	Port
PC0 → Switch0	Fa0/1
PC1 → Switch0	Fa0/2
PC2 → Switch0	Fa0/3
PC3 → Switch1	Fa0/1
PC4 → Switch1	Fa0/2
PC5 → Switch1	Fa0/3
Switch0 → Switch1 (uplink)	Fa0/24 ↔ Fa0/24

5. IP ADDRESS CONFIGURATION

Device	IP Address	Subnet Mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0
PC2	192.168.1.3	255.255.255.0
PC3	192.168.1.4	255.255.255.0

PC4	192.168.1.5	255.255.255.0
PC5	192.168.1.6	255.255.255.0

All devices share the 192.168.1.0/24 subnet. No default gateway required.

6. PROCEDURE

1. Open Cisco Packet Tracer. Place 2 × Cisco 2960 switches and 6 × PCs on the workspace.
2. Connect each PC to its switch using Copper Straight-Through cables (ports as listed above). Connect Switch0 Fa0/24 → Switch1 Fa0/24 for the uplink.
3. For each PC: click PC → Desktop → IP Configuration → Static. Enter the IP address and subnet mask from the table above.

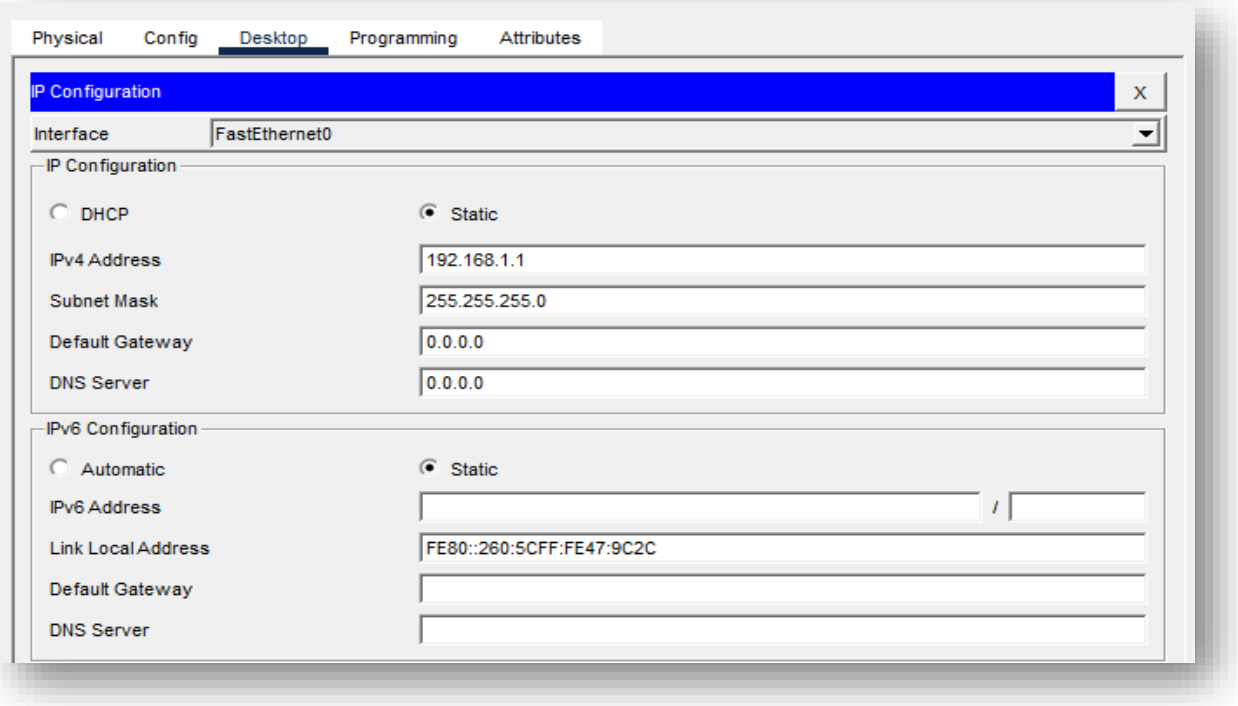


Figure 1: End PCs IP Address Configuration

4. Wait for all link LEDs to turn green (STP convergence). Switch to Simulation Mode (bottom-right). Ensure ICMP is enabled in Event List filters.
5. Select Add Simple PDU (envelope icon). Click PC5 as source, PC1 as destination. Press Play to run the simulation.

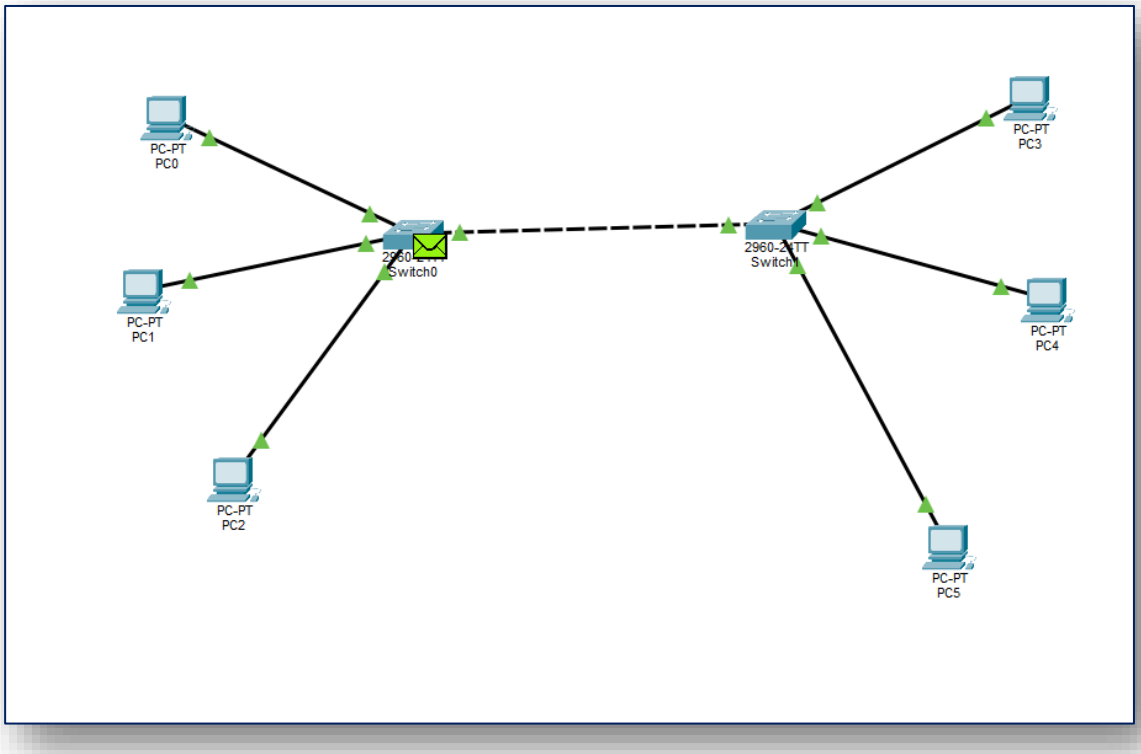


Figure 2: Stimulation on the Network Designed

7. RESULTS

Packet flow observed in the Simulation Event List:

Time (s)	Last Device	At Device	Type
0.000	PC5	Switch1	ICMP
0.001	Switch1	Switch0	ICMP

0.002	Switch0	PC1	ICMP
0.003	PC1	Switch0	ICMP
0.004	Switch0	Switch1	ICMP
0.005	Switch1	PC5	ICMP

PC5 → Switch1 → Switch0 → PC1 → Switch0 → Switch1 → PC5

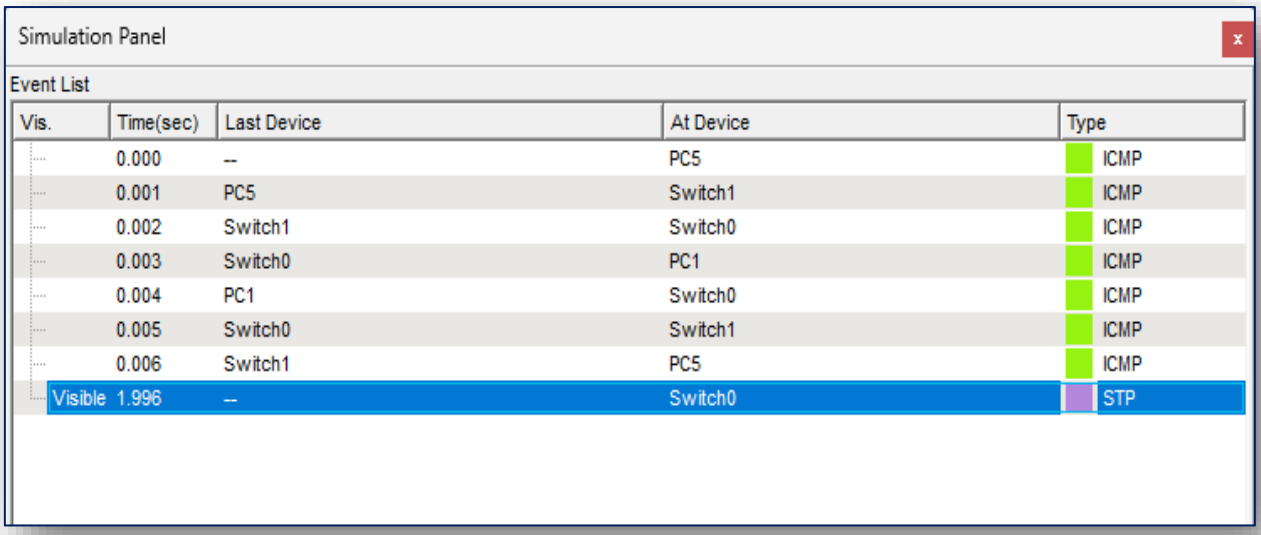


Figure 3: Simulation Event List showing all ICMP entries above.

8. CONCLUSION

Two Cisco 2960 switches were successfully interconnected and six PCs were configured within the 192.168.1.0/24 subnet. ICMP testing in Simulation Mode confirmed full bidirectional connectivity across both switches, demonstrating that Layer-2 switching enables seamless LAN communication without requiring a router.